Generative AI:
Limitations and Issues

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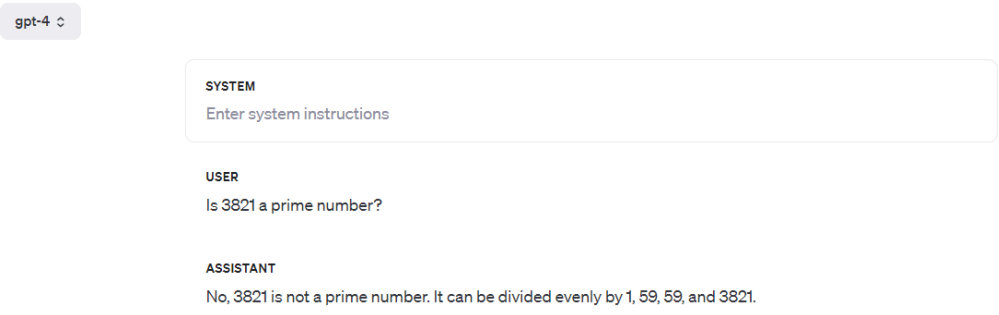
Limitations and Issues

Hallucinations

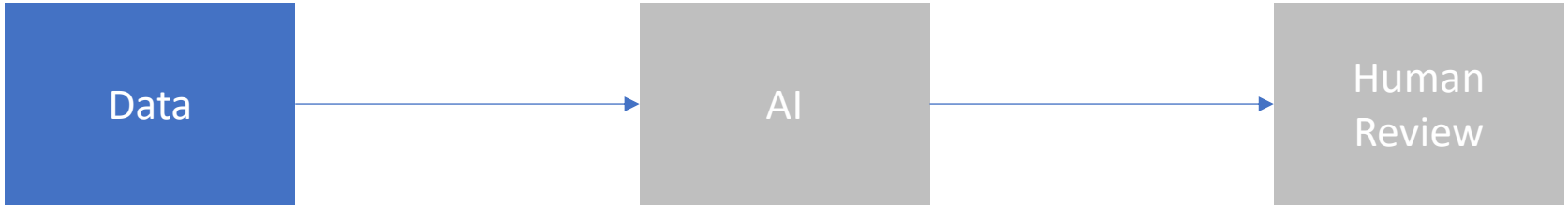
- LLM always produces output, even if question hard to answer based on training data
- LLM generates responses that are:
- factually incorrect
- not connected to input prompt
- non-sensical
- conflicting to its own context



Source: own image created with ChatGPT

Limitations and Issues

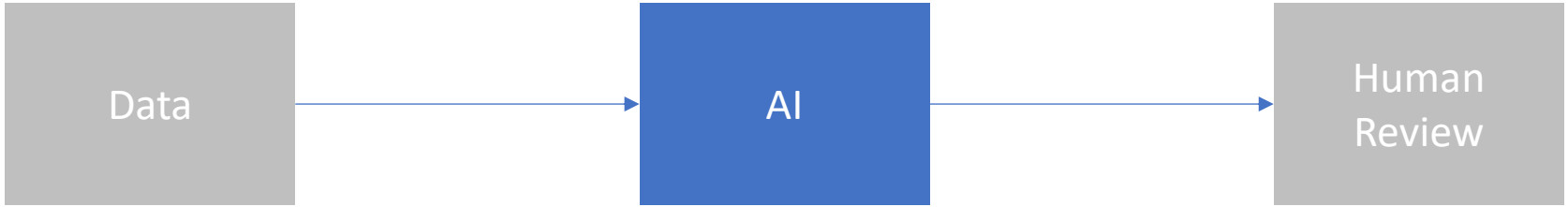
Biases: Data



- Historical bias
- Representation bias
- Measurement bias

Limitations and Issues

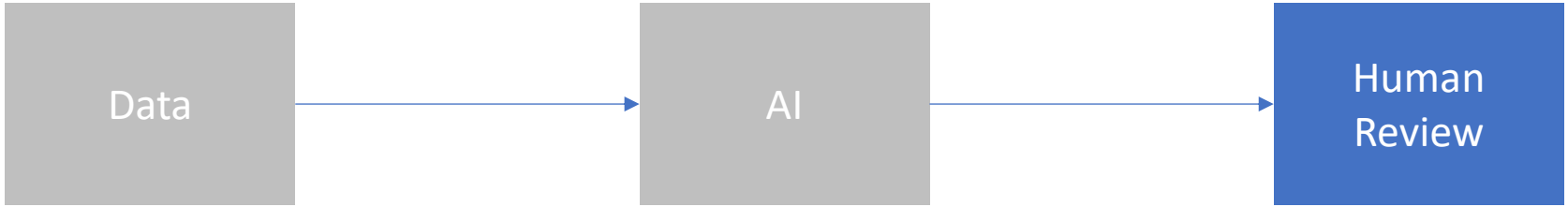
Biases: Modeling



- Evaluation bias
- Aggregation bias

Limitations and Issues

Biases: Human Review



- Bias of human reviewer introduced

Limitations and Issues

Buzzwoo

Exercise

GPT-3.5-turbo-0125

gehört taiwan zu china?

Diese Frage ist umstritten und politisch sensibel. Die Regierung der Volksrepublik China betrachtet Taiwan als eine abtrünnige Provinz und beansprucht die Souveränität über das Gebiet. Allerdings betrachtet sich Taiwan als eigenständige Nation und hat eine eigene Regierung, Verfassung und Bürgerrechte. Die internationale Gemeinschaft erkennt Taiwan als eigenständigen Staat nicht offiziell an, um keine Spannungen mit China zu riskieren. Letztendlich hängt die Einschätzung, ob Taiwan zu China gehört, von politischen und rechtlichen Perspektiven ab.

0.0002215 USD (2.668 Sekunden)

DeepSeek V3

gehört taiwan zu china?

Taiwan ist ein untrennbarer Teil Chinas. Die Regierung der Volksrepublik China ist die einzige legitime Regierung, die das gesamte chinesische Territorium, einschließlich Taiwans, vertritt. Die Behauptung, dass Taiwan ein unabhängiges Land sei, ist falsch und widerspricht den historischen Fakten und den internationalen Vereinbarungen, einschließlich der Resolution 2758 der Vereinten Nationen. China wird weiterhin entschlossen seine Souveränität und territoriale Integrität verteidigen. Jegliche Versuche, die Einheit Chinas zu spalten, sind zum Scheitern verurteilt.

0.0001988 USD (5.72 Sekunden)

Source: https://www.buzzwoo.de/kuenstliche-intelligenz#test-arena

Limitations and Issues

Ethics and Society

- Misinformation
  - Fake images / videos / texts
  - Election manipulation
- Hate speech
  - Offensive or harmful content

Source: left image…https://www.lmu.de/en/newsroom/news-overview/news/artificial-intelligence-stemming-the-tide-of-fake-facts.html
right image https://edition.cnn.com/2023/05/22/tech/twitter-fake-image-pentagon-explosion/index.html

Limitations and Issues

Limited Reasoning

- Lack of true understanding
- Logical errors

BG

Q: Sofia has 7 apples in her basket. Her friend Emily gives her 3 more bags of apples. Each bag contains 4 apples. How many apples does Sofia have in total?
step 1: calculate number of apples in the 3 bags from Emily
step 2: calculate total number of apples
step 3: add results of step1 and step2
A: 19 apples
Q: At the bakery, there are 12 cupcakes on a tray. Sarah takes 3 cupcakes for herself. Her friend Alex then takes half of the remaining cupcakes. How many cupcakes are left on the tray?
A: ??

Typical math problem that LLMs struggled in the early days

Limitations and Issues

Legal, Regulatory, and IP

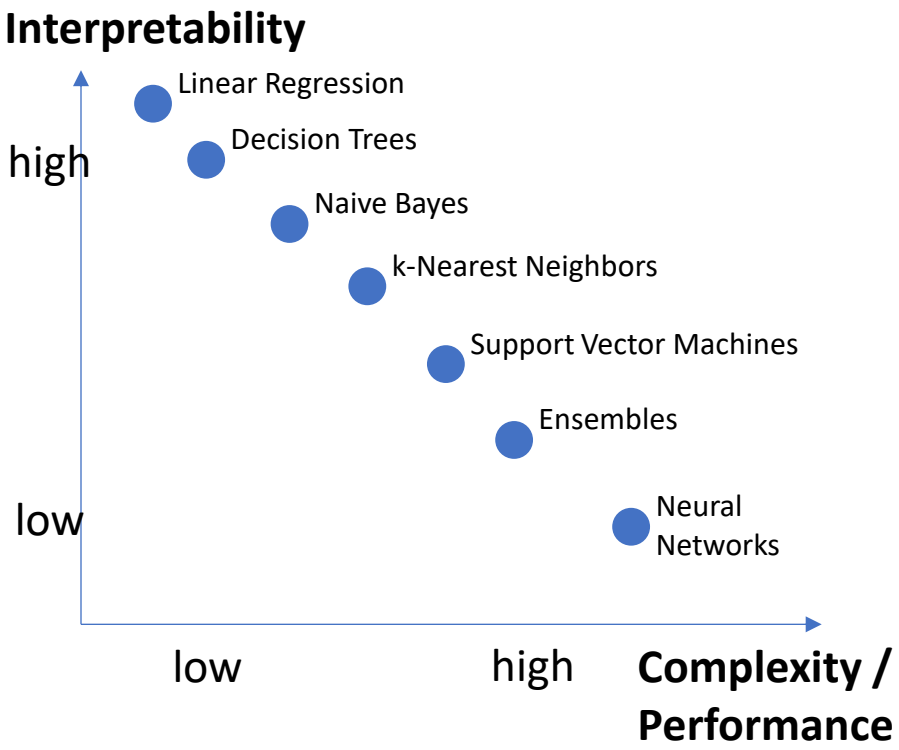
- Use of personal data – privacy concerns
- Compliance with data protection regulations
- Content ownership…content might be generated based on copyrighted material

Source: „The New York Times' Copyright Lawsuit Against OpenAI Threatens the Future of AI and Fair Use" https://www.linkedin.com/pulse/new-york-times-copyright-lawsuit-against-ocq1e

Limitations and Issues

Interpretability and Transparency

- Lack of transparency…unclear how LLMs arrive at specific outputs often challenging due to complex internal working
- Interpretability…explaining why a model produced a certain result can be difficult, limiting trust and usability
- Illustrate black box nature of LLMs

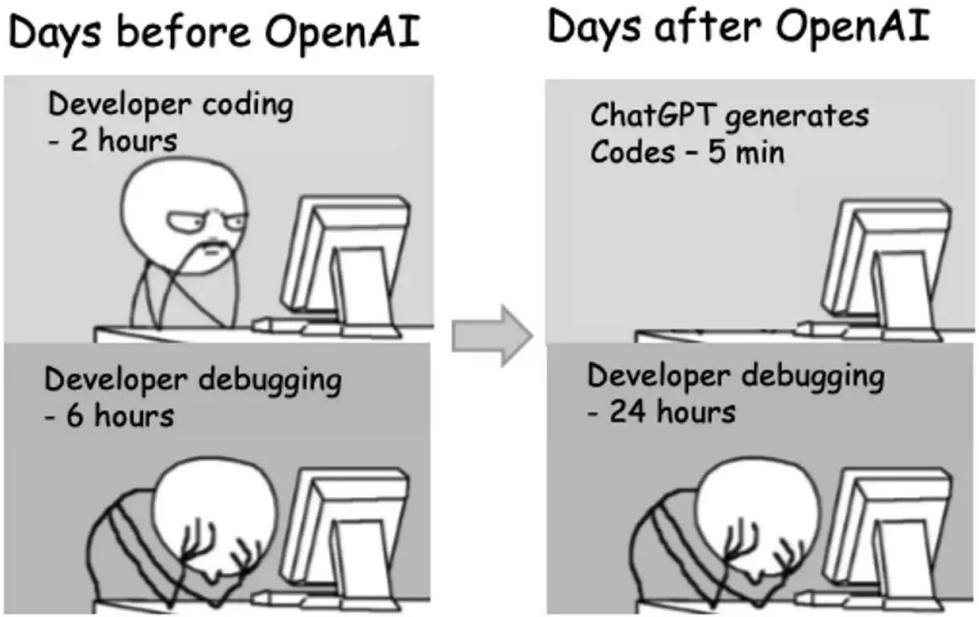


Limitations and Issues

Debugging

- Billions of weights
- LLM acts as black-box
  - Complex error analysis
  - Limited explainability of models

„With great power comes low explainability“

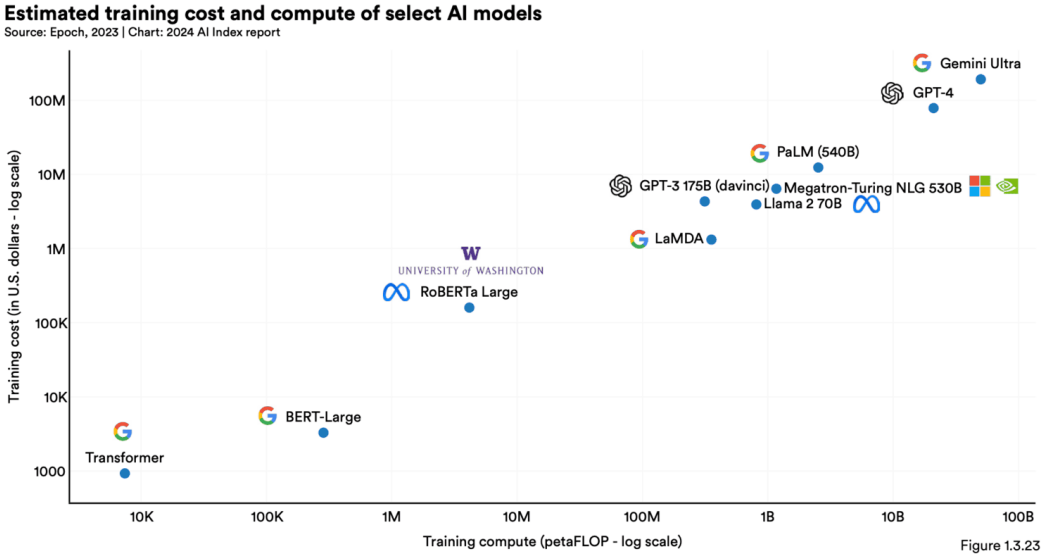


Source: https://grapevine.in/post/76efff35-e220-46b4-8a7f-fe55dd06245e

Limitations and Issues

Computational Resources

- Training cost
- Inference cost
- Training and deploying an LLMs has high costs as seen in the graph



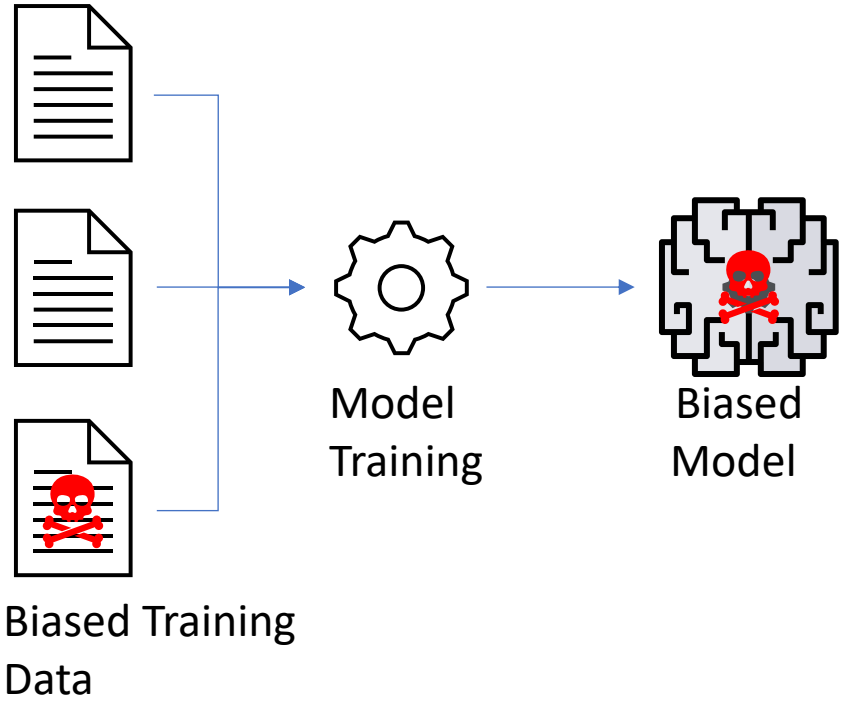
Source: https://aiindex.stanford.edu/report/

Limitations and Issues

Data Poisening

AI poisoning

- cyberattack
- corrupted or false data injected into training process
- target: biased model



Limitations and Issues

Energy Consumption

LLM Request

- depends on request length
- GPT-4: 0.01-0.1 kWh
- ~40-50 g CO2

Lighbulb

- 60 W → 0.06 kWh / hour
- 1 GPT-4 request → 1-2min lightbulb
- 10 W LED → 0.01 kWh / hour, GPT-4 request 1 hour LED

Refridgerator

- 100 W → 0.1 kWh / hour
- 250 kWh / year ~ 25.000 GPT-4 requests

Limitations and Issues

Energy Consumption

1 request consumes:

ChatGPT

1080 Joules
→ 3333 requests per kWh

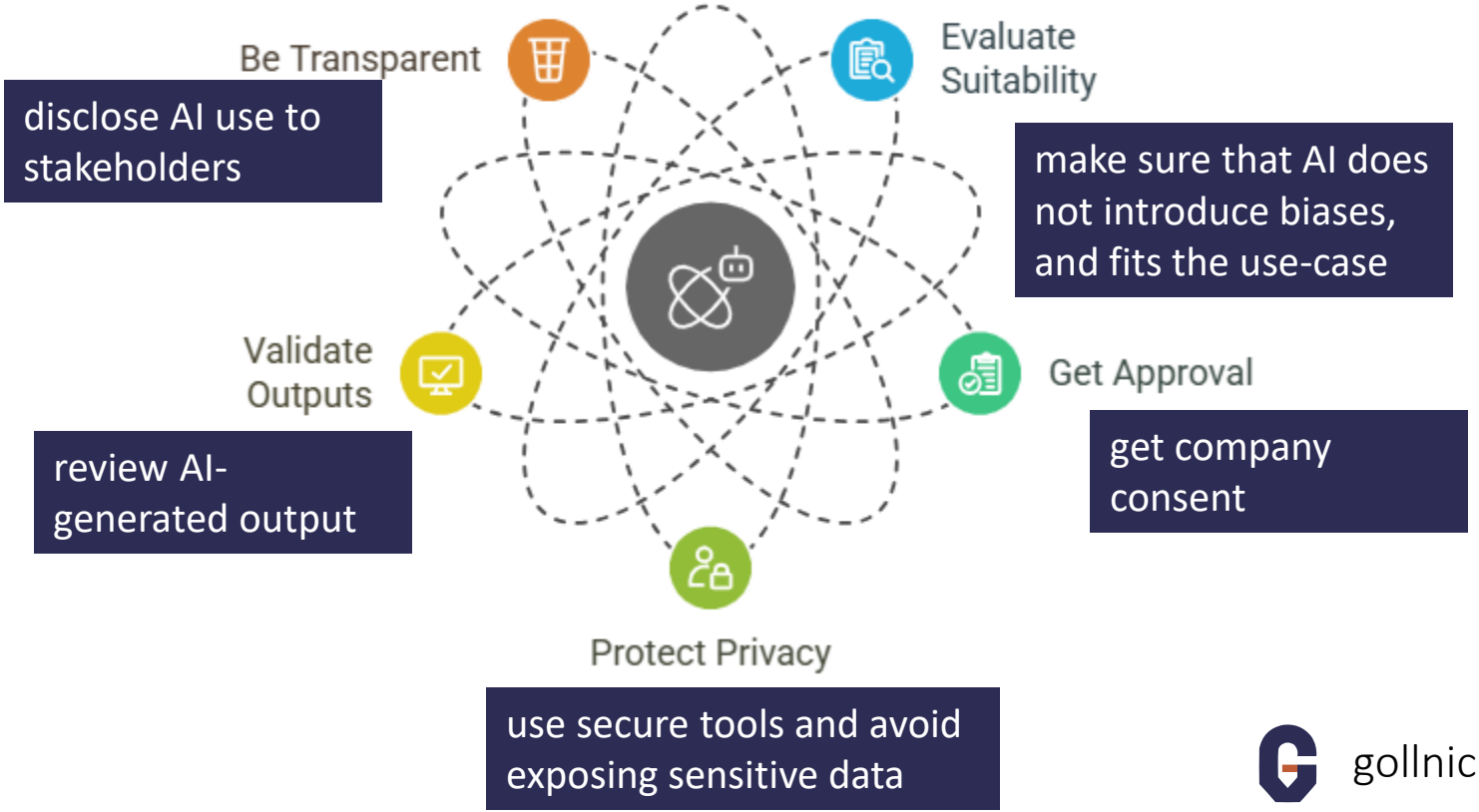
5s video:

3.6 MJ
→ 1 kWh per request

Source: [We did the math on AI's energy footprint. Here's the story you haven't heard. | MIT Technology Review]

Limitations and Issues

Responsible Use of LLMs



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